

Olivia Chen

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Education

Merivale High School

International Baccalaureate (IB) Diploma and Ontario Secondary School Diploma (OSSD)

GPA: 97/100, 99/100 Graduating Top 6; IB Score: 43/45

Ottawa, Ontario

September 2021 — June 2025

Projects

Bookd 📖

- Developed and shipped a cross-platform football social media application using **React Native** and **Expo** with an end-to-end serverless backend using **Firebase**. Reached **5,000+** downloads within two months of release and achieved a **13-minute** average engagement time and **25%** D30 retention rate. Featured in the **Ottawa Business Journal** 📰.
- Architected a database spanning **20+** collections on **Cloud Firestore**, housing the largest commercially available football repository (**100,000+** matches with associated data) alongside user-generated content and social graph data. Designed for horizontal scale by implementing denormalised aggregates, composite indexes and atomic counter operations.
- Designed a scalable social graph and feed system with social primitives (likes, comments, replies, reposts, follows) across distinct contexts (reviews, lists, match discussions, team chats) using cursor-paginated infinite queries, denormalised hot-path snapshots, inverted indexes and event-driven **Cloud Function** triggers for counter maintenance and cascading cleanup.
- Engineered a two-phase projection-then-reconcile pattern with a half-hourly **Cloud Scheduler** job to work around **API Football's 1-hour update lag**, maintaining the real-time accuracy of **50+** leagues' standings.
- Built privacy and content controls including bidirectional user blocking with cascade-filtering through all queries, configurable spoiler modes, a content moderation pipeline using **Google Cloud Vision API** and text profanity detection, **Klipy**-backed GIF embeds, account deletion with full cascade cleanup and Instagram Stories sharing with on-device snapshot rendering.
- As one of two engineers on the project, owned product design (**Figma**), full-stack architecture, App Store/Play Store releases and OTA hotfix workflow for client-side incidents.

Flow 📄

- Built a real-time computer-vision pipeline allowing users to capture quotes by tracing them with a finger by tracking the index fingertip via **MediaPipe** hand-landmark detection and running **OCR** on the traced region to retrieve the quote.
- Trained a gesture classifier in **PyTorch** on fingertip-motion data to distinguish intentional tracing from idle movement and benchmarked it against a rule-based baseline to justify the design choice.
- Integrated a local **Whisper** speech-to-text model to transcribe the reader's spoken commentary, persisting each scanned quote with users' relevant spoken thoughts in a **PostgreSQL** database

Experience

Nokia

Ottawa, Ontario

Software Engineering Intern

July 2024 — August 2024

- Led and built a **PowerApps** application tracking **global network outages and maintenance** for Nokia's global leads. Integrated **SharePoint** to streamline entries and **Power BI** for log visualisation and insights.
- Optimised queries by indexing high-traffic columns to work around PowerApps's **500-record delegation limit**

Mayer Research Group — University of Ottawa

Ottawa, Ontario

Research Assistant

June 2025 — September 2025

- Co-authored "Probing Metal Ion Interactions in Stacked Polycyclic Aromatic Hydrocarbons as a Molecular Model for Superconductivity," a paper published in the peer-reviewed **University of Ottawa Science Undergraduate Research Journal (OSURJ)**, under the supervision of Professor Paul Mayer.
- Employed **Gaussian** to calculate binding energies corresponding to mechanistic pathways deduced from **mass spectrometry** results.

Technical Skills

Languages: Python, Swift, JavaScript/TypeScript, SQL

Frameworks/Services: Firebase, PostgreSQL, PyTorch, MediaPipe, Whisper, SwiftUI, React, React Native, Selenium

Tools: Git/Github, VSCode, Xcode, Expo, Vercel, Figma, Gaussian, Tesseract

Awards

- Greville Smith Scholarship, McGill's flagship scholarship modeled after the Rhodes Scholarship (**1 of 6 out of 7,000**)—\$48,000
- Samsung Solve for Tomorrow regional finalist (**top 12 nationwide**)—\$2,500
- Invited to speak at the **Canadian Society for Mass Spectrometry** Student Conference at Trent University